

Automotive Market Analysis

Vladyslav Yakymchuk

Data Source & License

Source: Kaggle — Vehicle Sales Dataset



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All conclusions are based on the provided dataset and may not reflect real-world market conditions.

Data description:

- **Size:** ~500,000 vehicle listings
- **Country:** USA
- **Time span:** December 2014 - July 2015

Presentation Agenda



Section 1: Market Structure

- Sales by body type
- Top 20 brands by sales volume
- Transmission type distribution



Section 2: Pricing

- Price dependency on mileage
- Pricing by year of manufacture
- Average prices by body type



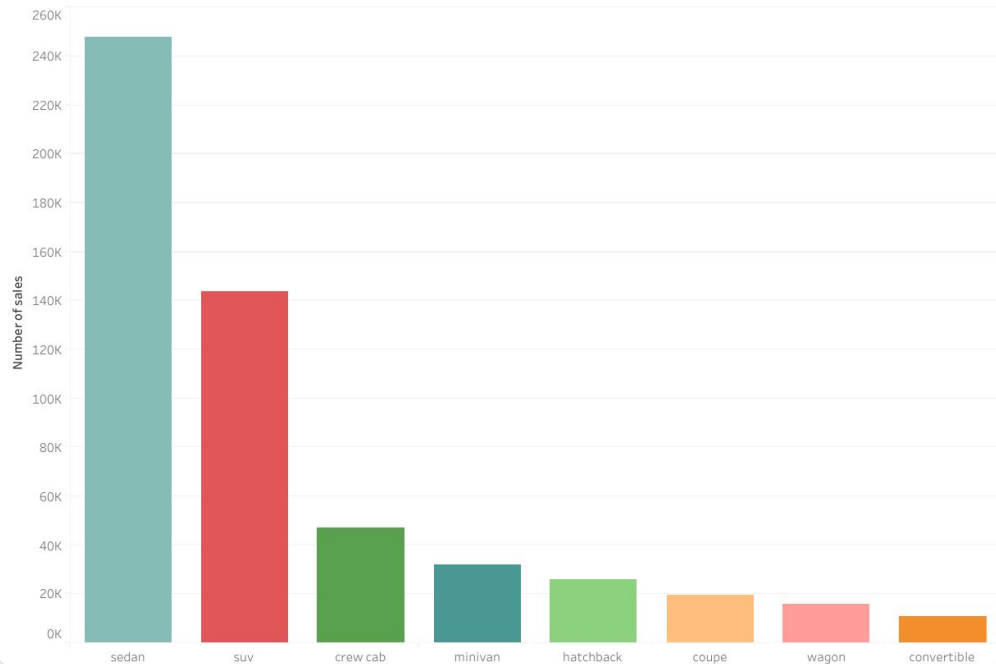
Section 3: Market Segmentation

- Most affordable brands
- Most expensive brands
- Popular car colors

Sales Structure by Body Type

Section 1: Market Structure

Number of sales by car body



Key Findings:

Sedan — clear market leader (~250K sales)

SUV — second place (~145K sales)

Crew cab — third place (~48K sales)

Other types (minivans, hatchbacks, coupes, wagons, convertibles) have lower volumes

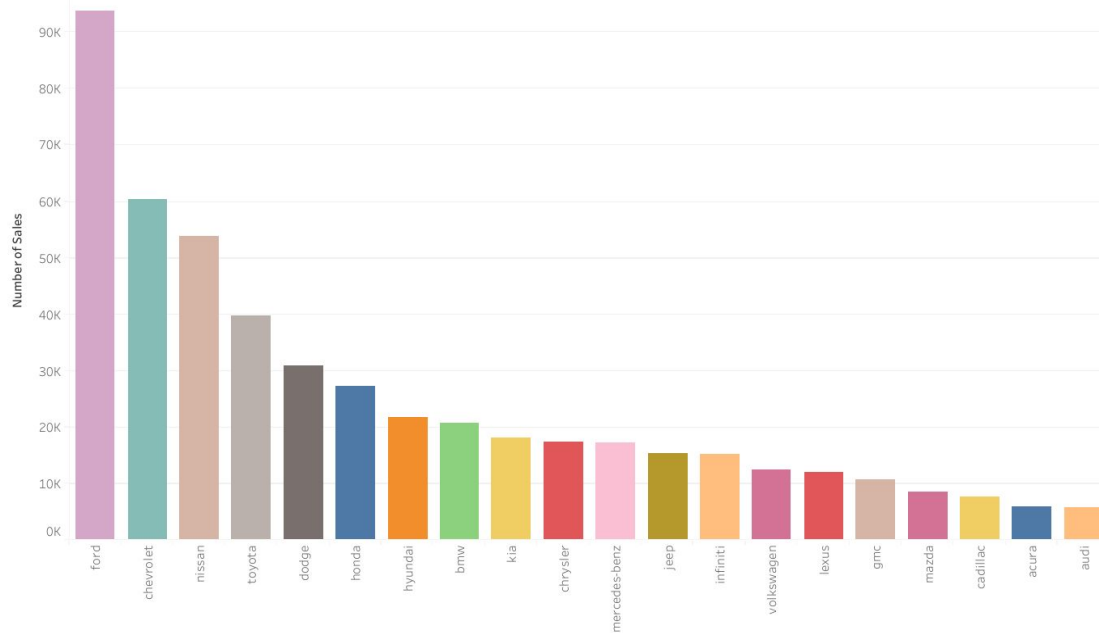


Insight: Sedans and SUVs account for over 70% of the entire market. These segments are priority for investment and marketing.

Top 20 Brands by Sales Volume

Section 1: Market Structure

Top 20 car brands by sales number



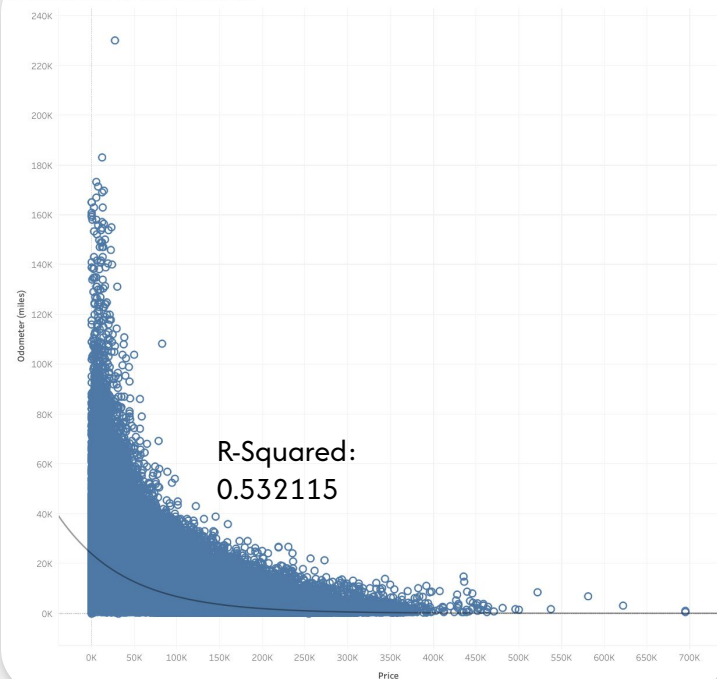
Market Leaders:

1. **Ford** — ~94K sales
2. **Chevrolet** — ~60K sales
3. **Nissan** — ~54K sales
4. **Toyota** — ~40K sales
5. **Dodge** — ~31K sales
6. **Honda** — ~27K sales
7. **Hyundai** — ~22K sales
8. **BMW** — ~21K sales
9. **Kia** — ~18K sales
10. **Chrysler** — ~17K sales

Price Dependency on Mileage

Section 2: Pricing

Dependence of car price on mileage



Depreciation Trend:

Price decays exponentially with mileage, not linearly — the steepest drop happens in the first ~50,000 miles.

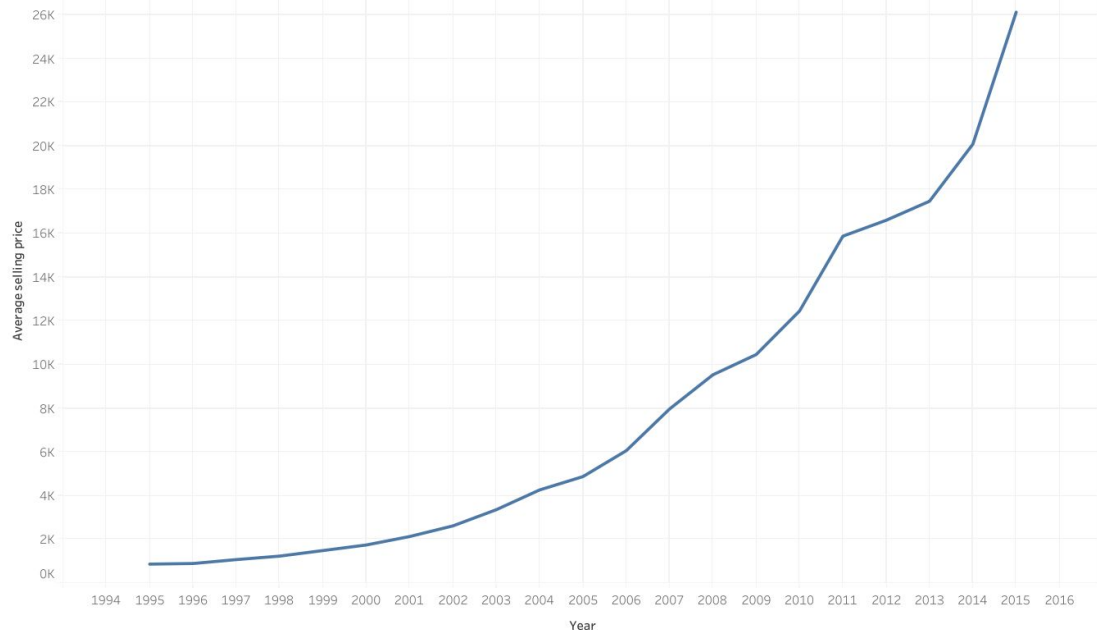
$R^2 \approx 0.53$ (exponential fit, raw listings)

Note: R^2 of 0.53 reflects individual listings — mileage explains roughly half the price variation; the rest comes from make, condition, and year.

Pricing by Year of Manufacture

Section 2: Pricing

Average selling price of cars by year of manufacture



Price Dynamics:

1995-2005: Slow growth from \$1K to \$5K

2005-2010: Accelerated growth to \$11K

2010-2015: Exponential growth to \$20K



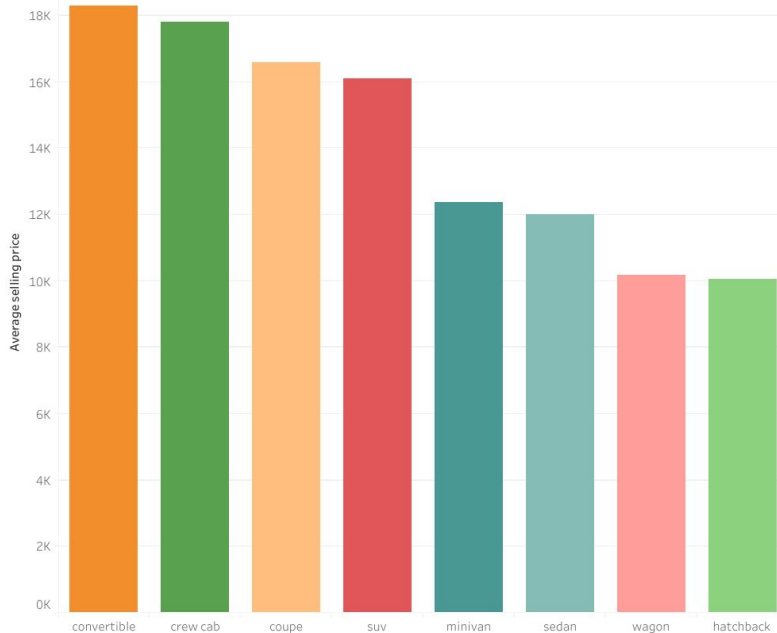
Insight: Cars newer than 5 years retain the most value. After 10 years, depreciation slows down.

Formula: Each year adds ~\$1,000-2,000 to value for newer cars (2010+)

Average Prices by Body Type

Section 2: Pricing

Average price for each body type



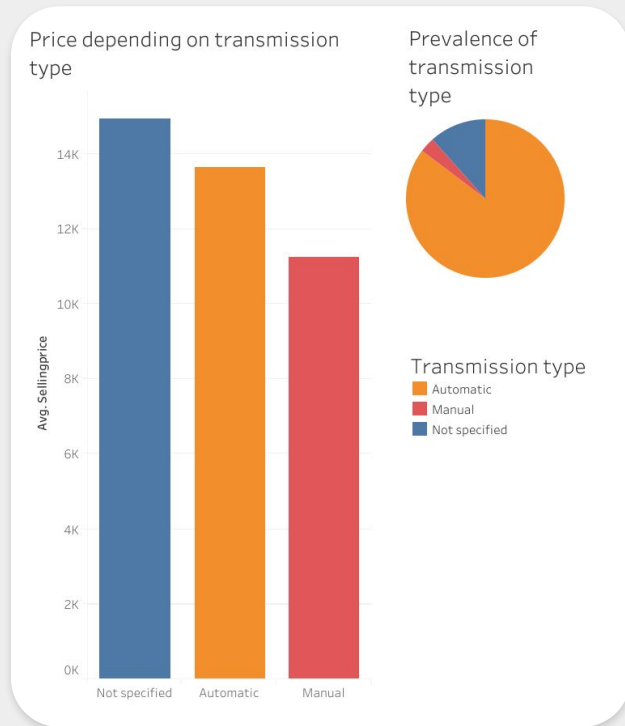
Price Ranking:

1. **Convertible** ~\$18,500
2. **Crew cab** ~\$17,900
3. **Coupe** ~\$16,800
4. **SUV** ~\$16,200
5. **Minivan** ~\$12,500
6. **Sedan** ~\$12,000
7. **Wagon** ~\$10,300
8. **Hatchback** ~\$10,000

 **Conclusion:** Sports and premium types are 50-80% more expensive than practical ones

Transmission Type Distribution

Section 2: Pricing



⚙️ Transmission Prevalence:

Automatic — dominates the market (over 80%)

Manual — small share (up to 10%)

Not specified — around 5-10%

⚙️ Transmission Price:

Not specified — the most expensive option (probably electric cars or other transmission types, such as robotic)

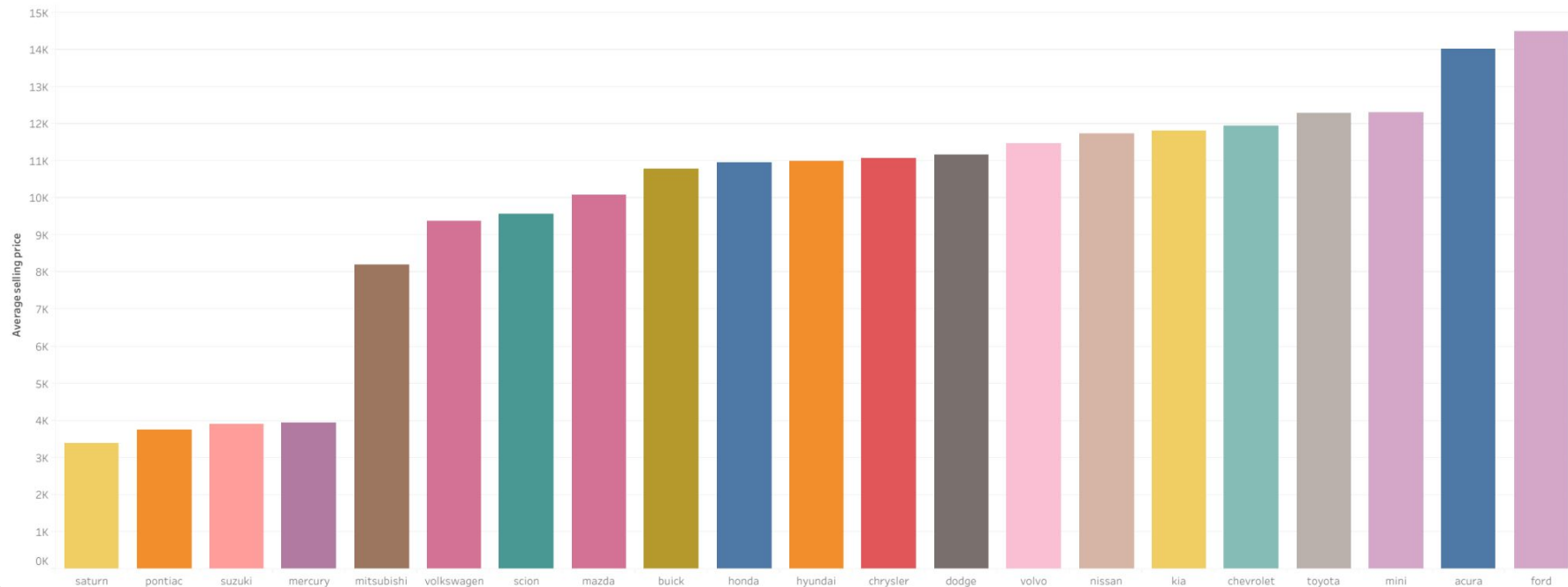
Automatic — on average \$13.5K

Manual — \$11K on average, the cheapest type of transmission

Most Affordable Brands

Section 3: Market Segmentation

Top 21 cheapest car brands (with at least 1,000 sales)



Most Affordable Brands

Section 3: Market Segmentation

Top 5 Cheapest:

1. Saturn (~\$3,500)
2. Pontiac (~\$3,800)
3. Suzuki (~\$4,000)
4. Mercury (~\$4,000)
5. Mitsubishi (~\$8,200)

Mid-range Segment:

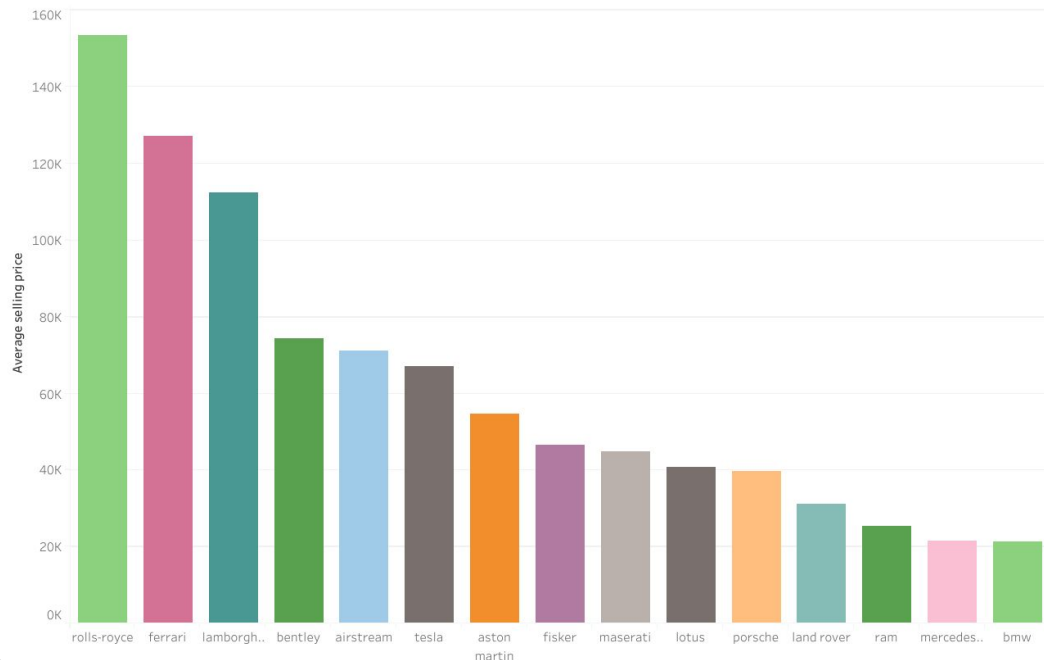
- Volkswagen (~\$9,500)
- Scion (~\$9,500)
- Mazda (~\$10,000)
- Buick (~\$10,800)
- Honda (~\$11,000)

 **Observation:** Saturn, Pontiac, Mercury — discontinued brands, hence low prices on the secondary market

Most Expensive Brands

Section 3: Market Segmentation

Top 15 most expensive car brands



Price Ranking:

1. Rolls-Royce ~\$156,000
2. Ferrari ~\$128,000
3. Lamborghini ~\$113,000
4. Bentley ~\$74,000
5. Airstream ~\$70,000

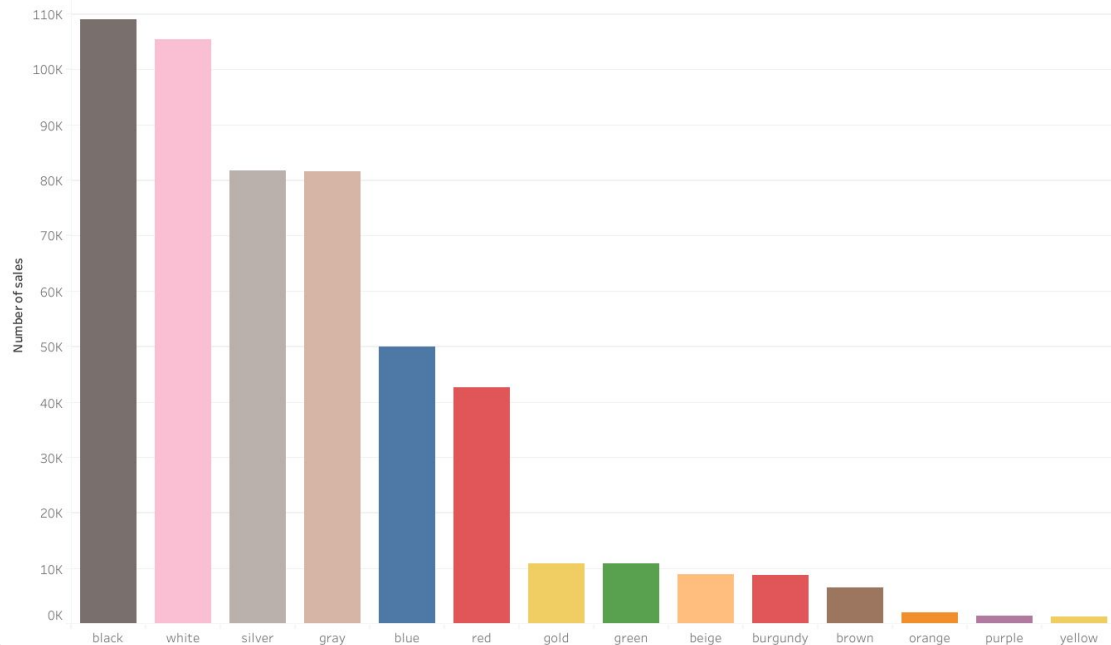
6. Tesla (~\$67K) • 7. Porsche (~\$55K) 8. Fisker (~\$47K) • 9. Maserati (~\$45K)

🔍 Analysis: Exclusive brands are 50-100× more expensive than budget ones. Rolls-Royce costs as much as ~45 Saturns.

Popular Car Colors

Section 3: Market Segmentation

Number of sales by car color



Top Colors:

1. **Black** — ~112K sales
2. **White** — ~108K sales
3. **Gray/Silver** — ~82K each
4. **Blue** — ~50K
5. **Red** — ~42K



Conclusion: Neutral colors (black, white, gray) account for 73% of the market

Key Takeaways



Market Structure:

Sedans and SUVs dominate (70%+ market share). Ford leads sales. Automatic transmission is standard (80%+).



Pricing:

Price drops 40% after 50K km mileage. Each year adds \$1-2K for newer cars. Convertibles and crew cabs are the most expensive types.



Segmentation:

Rolls-Royce (\$156K) vs Saturn (\$3.5K) — 44× difference. Black and white are most popular colors.



Recommendations:

- Focus on sedans and SUVs with automatic transmission
- Prioritize neutral colors (black, white, gray)
- Diversify budget and premium segments